

Systems & Materiel Engineering

Master of Engineering

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PROGRAM DESCRIPTION

The mission of the Master of Engineering in Systems and Materiel Engineering program is to provide students with enhanced opportunities for professional development and career advancement and provide employers with better skilled, more adaptable, and satisfied career employees.

It is an interdisciplinary professional master's degree program that focuses on creating processes and strategies for developing and managing complex systems over their lifecycles. The program involves the analysis, design, development, integration, test and evaluation, and disposal of complex operational systems.

ADMISSION REQUIREMENTS

The professional degree of Master of Engineering in Systems and Materiel Engineering program is open to qualified applicants holding a regionally accredited baccalaureate degree in engineering, science, mathematics, management, or business. Students without an undergraduate degree in engineering will be required to successfully complete GEN 500: Engineering Systems Analysis.

Please see the general graduate admission requirements [here](#). Specific requirements to this program are as follows:

1. Applicants must have an overall Grade Point Average (GPA) of at least 3.00 on a scale of 4.00, or have passed the Fundamentals of Engineering (FE) Examination.
2. Three letters of recommendation from individuals best suited to assess the applicant's quality of academic or professional performance. The letters must testify to the applicant's ability and motivation to succeed in the program. Letters from friends and family members are not acceptable.
3. One-page statement of intent should outline personal career goals and reasons for interest in the Systems and Materiel Engineering program at AAMU. This statement should include information about how the Master of Engineering degree in Systems and Materiel Engineering program may help in achieving them.
4. Resume/Curriculum Vitae.

DEGREE REQUIREMENTS

The Master of Engineering degree in Systems and Materiel Engineering is a professional degree and does not require a thesis, but requires a capstone project. The program requires a minimum of 30 credit hours of graduate-level courses with a cumulative grade-point-average of 3.0. Students may, upon departmental approval, transfer a maximum of twelve credit hours of approved graduate credits from an accredited institution.

(MAT) Systems & Materiel Engineering – Non-Thesis[#]

30-33 Credit Hours

MinGPA cumulative 3.0. MinGrade C*. Degree M.Eng.

CORE COURSES	
¹ GEN 500 Engineering System Analysis	3
GEN 601 Life-Cycle Design Engineering	3
GEN 603 Analysis and Simulation Methods	3
GEN 604 Test and Evaluation Engineering	3
GEN 605 Engineering Project Management [RPP]	3
GEN 606 Fund of Systems Engineering	3
GEN 690 Materiel Engineering Project	3
ELECTIVES	12
COMPREHENSIVE EXAM	
Grade is Pass / Fail.	
Exam composed jointly by Advisory Committee.	
To be taken after ≥ 66% completion of required coursework.	

[#] Offered online.

*One grade of C allowed at graduation.

[RPP] = Research project paper or portfolio required.

¹ Required if undergraduate degree is not in engineering. MinGrd of B required.

Concentrations, Specializations & Electives

(MAT) GENERAL ENGINEERING ELECTIVES	
CE 501 Structural Steel Design	3
CE 502 Reinforced Concrete Design	3
CE 504 Hydraulic Engineering & Design	3
CE 508 Foundation Design	3
CE 510 Transportation Engineering & Design	3
CE 512 Pavement Systems	3
CE 555 Wastewater Treatment	3

(MAT) ELECTRICAL ENGINEERING ELECTIVES	
EE 503 Feedback System Analysis & Design	3
EE 504 Communication Theory	3
EE 510 Microwave Engineering	3
EE 513 Rocket Propulsion	3
EE 520 Power Systems I	3
EE 525 High Performance Computing/Networks	3
EE 531 Advanced Semiconductor Engineering	3
EE 541 Digital Signal Processing	3
EE 551 Integrated Circuit Fabrication	3
EE 552 Semiconductor Instrumentation	3

(MAT) MECHANICAL ENGINEERING ELECTIVES	
ME 512 Analysis and Synthesis of Gas Turbines and Components	3
ME 513 Rocket Propulsion	3
ME 516 Gas Dynamics	3
ME 541 Renewable Energy	3
ME 542 Solar Thermal Engineering	3
ME 572 Economic Evaluation of Design	3
ME 581 Quality and Reliability Assurance	3
ME 582 Operations Planning and Scheduling	3